PETU LABS — TrafficFlowBench Award Paper (draft, content stage)

2026 IEEE Big Data Cup: TrafficFlowBench

Team: Petu-Labs · Kaggle: adityapaagare

Candidate: 0.85330 floor (Kaggle ref 57429904) · `submissions/submission_w57_d1d2_v1.csv`

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md5 e6db2a22a5e676955ab225358c0e378c · 6,980,503 rows

> *Draft status: content draft for the 10-page award PDF (code + PDF due 2026-11-10 23:59 AoE). Rewritten 2026-11-06 to the certified floor (supersedes the 0.81787 / ref 57065852 draft of 2026-10-12, retained in version history). Every numbered claim below is backed by a named artifact in the award package; the reproducibility section names the exact acceptance tests the organizers' rerun must satisfy. Claim labels follow the campaign constitution §3.1 (FACT—CAMPAIGN RECORD, MEASURED, REPRODUCED, INFERENCE, CONTRADICTED/FAILED, HUMAN DECISION, UNKNOWN).*

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1. Abstract

We present the TrafficFlowBench solution of team Petu-Labs (public validation score **0.85330**, Kaggle submission ref 57429904), built under one organizing principle: **optimize expected rank on the hidden private month, not the observed validation month.** The competition's public leaderboard scores a single validation month; final standings are decided by a different, independently generated hidden month. Every component of our pipeline was therefore gated on multi-month stability before promotion: a candidate that could not show same-sign gains on at least three of five distinct months was killed or banked, never shipped.

The submitted bundle is the certified floor: our 0.84994 parent bundle (ref 57369480) with **exactly two changes** — W57 split-fidelity fixes D1 and D2, which re-estimate two serving-time statistics from each scored split's own released data instead of pooled training data. Its T1 leg is the W28 scaled gradient-boosting ensemble (two CatBoost regressors — speed, flow — × two seeds, trained on 3,725,267 rows with masking augmentation matched to the observed masking census, 20/30/50%); its T2 leg is the J1 queue construction with W54-A onset; its T4 leg is the W54-C own-counts refit. The floor was certified by a dedicated audit (W58-L1, 2026-10-20): byte-identical to the submitted file, composition attributed by construction, level discipline PASS (zero train-pooled level carryover), class composition 6 target-identified / 17 seed-invariant / 12 fitted-residue pairs. The award package reproduces the submission from raw inputs with **prediction-equality acceptance** (per-cell fidelity ≤ 1e-14 on speed, exact on flow and both non-state legs), and an independent full-population retrain replicated the W28 training procedure to |ΔS_state| = 0.000524 (RS1R Stage 1, 2026-11-04). The honest record behind the floor is 253 registered experiments, six final doors closed with named mechanisms, and a kill record that now measures — rather than asserts — that only retraining beats the fitted surfaces. The full rehearsal, environment pins, and frozen-artifact manifest are documented in §8.

2. Introduction

TrafficFlowBench asks for a full traffic state of a motorway network from masked sensor observations: per-link 5-minute speed and flow (T1), queue onset flags (T2), physics consistency of the predicted field (T3), and origin–destination matrix estimation (T4). The official score is

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S_total = 0.35·S_state + 0.30·S_queue + 0.15·S_physics + 0.20·S_ODME
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with S_state = 0.54·S_speed + 0.46·S_flow. The public leaderboard reflects one validation month; the organizers state that final standings use a different hidden month. This single fact shaped our entire campaign: **a gain that**

exists only on the validation month is a liability, not an asset. Our promotion contract therefore required cross-month stability (same-sign gains on ≥ 3 of 5 distinct months: four blinded months plus a sealed February selection month), a deflated significance bar (2.67σ, calibrated to ~150 lifetime trials so that luck alone cannot promote), and a bundle ablation that drops any component whose marginal is below +0.002 total-score units. No gate was recalibrated after seeing results; the one uniform gate change in campaign history (K3 → K3′, 3 November) is disclosed as post-result in §6 and priced accordingly — the floor ships alongside, never instead of, the protection it provides.

3. Data

The released package covers ten motorway panels (D7_I10, D7_I210, D7_I405, D12_I5, D12_I405, each direction) with masked mainline state observations at 5-minute resolution, a road-network graph (link lengths, lane counts), and a sample submission key (6,980,503 rows). Training months available to us ran June 2030 – January 2031; February 2031 was sealed and was opened exactly once per experiment under a pre-registered frozen protocol (hashes recorded before opening, no post-open tuning). The observed masking census is approximately 20/30/50% per regime, and our training augmentation reproduces that census exactly (see §5.1). No external dataset, pretrained model, or paid API was used anywhere in the pipeline (README_EXTERNAL_DATA.md: none in any category).

4. Solution architecture

The submission is a four-leg bundle. Relative to our production baseline (ref 56788939, 0.78806), the floor carries the W28 T1 ensemble, the W54-C T4 refit, the W54-A onset leg, and the D1/D2 split-fidelity fixes:

Leg	Task	Content in the submission	Changed vs production?
T1	**state (speed/flow)**	**W28 scale CatBoost ensemble (§5) + D1 per-split priors (§5.5)**	**yes — ensemble + split priors**
T2	**queue**	**J1 ongoing (model_J1.txt, thresholds) + D2 per-split occ_ff (§5.5); W54-A onset rows**	**yes — D2 split occ_ff; onset verbatim**
T3	**physics consistency**	**conservation-guard adjudicated (§6)**	**input field only**
T4	**ODME**	**W54-C own-counts refit (odme_candidate_owncounts.csv)**	**yes — own-counts refit**

The integrity proof (W58-L1 audit + W57 row-diff report) establishes against the 0.84994 parent (md5 f596f2089403c552f38fd6bd0f9808a8): identical row count, submission_id, and task columns; all 70,708 ODME rows 0-diff; all onset rows 0-diff; state queue_pred/path_flow and queue speed_kmh/flow_vph/path_flow 0-diff; D1 moves 6,722,215 of 6,735,795 state rows (mean |Δ| 0.392 km/h speed, 10.32 vph flow); D2 flips 260 of 174,000 queue rows (41 $0\to1$, 219 $1\to0$). No full-leg swap was performed — the W56 assembly error mode is excluded by the replication proofs in §5.5. The floor is therefore a pure serving-time-statistics change by construction, and its score accounting is exact: the +0.00336 delta over the parent sits in the D1/D2 components (March read, class evidence only — §6).

5. The T1 leg: scaled CatBoost ensemble

5.1 Training data

build_train_w28_scale.py builds **3,725,267 rows in 10 shards** from the raw masked parquets of the eight training months (Jun 2030 – Jan 2031): 978,619 natural rows plus masking augmentation at 20/30/50% rates, matching the observed masking census. Per-shard md5s are recorded in builds/w28_t1_scale/data/shard_md5.txt (10/10 verified in the award rehearsal, re-verified 2026-11-06). The builder is memory-safe (one panel in memory at a time), deterministic given the raw inputs, and never touches February 2031.

5.2 Features

The v2 feature set (builds/w27_t1_scaleup/t1_features_v2.py, frozen list t1_feat_v2_scale.json) combines:

- **Episode/wave features**: congestion-wave and episode structure around

each cell, capturing the spatiotemporal context of the 5-minute grid.

- **Pooled priors** (priors_scale_{panel}.npz, ×10): empirical per-(link,

5-minute slot) mean speed and flow — deterministic sum/count accumulators built from 30 days pooled across the eight training months. They are **frozen inputs** to the pipeline (loaded verbatim at inference; regenerable from released data by the documented recipe), not fitted parameters. Their md5s are recorded in PRIORS_STATEMENT.md.

- Masked-observation statistics, link geometry (length, lanes), and

time-of-day/calendar encodings.

5.3 Model and training

Four models: speed/flow × seeds {42, 43}. Each is a CatBoostRegressor (depth 8, learning rate 0.05, up to 1500 iterations, l2_leaf_reg 3.0, early stopping on a 5% seed-deterministic validation split) trained on private Kaggle kernels from the on-Kaggle dataset adityapaagare/tfb-w28-scale (10/10 shard md5s verified in-kernel before training). The ensemble prediction is the **mean of the two seeds** per target. Model md5s are recorded in builds/w28_t1_scale/STATUS.json; the inference driver (infer_w28.py) verifies all four against the manifest before running. All four models were retrained from scratch on private kernels on 2026-10-12 with prediction-equality acceptance (train_meta identical to 16 digits, 2000-row probe predictions bit-identical, max|Δ| = 0.0 — EXECUTION_REPORT.md §2).

5.4 Inference and bundle assembly

infer_w28.py mode valpriv recomputes the v2 features from the raw masked parquets and writes 20 intermediate state files (cand_state_{split}_{panel}.csv, 10 panels × validation/private splits; unmatched=0 required per panel). merge_candidate.py performs the key-ordered merge with the verbatim T2/T4 legs into the 6,980,503-row submission. The pinned environment is requirements.txt (catboost==1.2.10, pandas==3.0.6, numpy==2.5.3, pyarrow==25.0.1, scipy==1.18.1, lightgbm==4.7.0, osqp==1.1.3, joblib==1.6.0; python 3.12).

5.5 The D1/D2 split-fidelity fixes (W57, 2026-10-20)

FACT — CAMPAIGN RECORD. The floor differs from its 0.84994 parent by exactly two component-level changes, built by the gated fix lane w57 (builds/w57_d1d2_splitfix/REPORT.md, authorization S34) and certified by the W58-L1 audit (intel/CERTIFIED_FLOOR_57429904.md):

- **D1 — T1 priors per scored split** (build_priors_split.py,

infer_d1.py). The shipped build_priors was called verbatim; the only change is the input population: instead of 30 pooled Jun–Jan train-split truth frames, the function received all masked day frames of the scored split itself (31 validation + 30 private days per panel). All 20 per-split priors: **0 fallback cells** (every link×slot cell has ≥1 observed cell in each split); 16,169,043 observed cells pooled. Inference ran through the shipped serve path (infer_w28.predict_day imported, models/ensemble/fallback/emitter unchanged); only the priors argument differed.

- **D2 — J1 occ_ff per scored split** (build_occff_split.py,

emit_d2.py). The shipped rule (median occupancy over masked cells with masked speed > 0.85·vf, min_samples 25) on all masked days of the scored split. **0 fallback links** in any panel×split. The shipped emitter produced the queue rows (frozen J1 model + thresholds); only the occ_ff table source changed.

REPRODUCED (replication proofs, the attribution basis). D1 base run with shipped priors vs the parent's state rows: max err speed **3.6e-15**, flow **9.1e-13** (float text round-trip only). D2 base run vs the shipped emitter files: **0 diffs** on all 174,000 rows. Because the base pipelines reproduce the parent exactly, every candidate-vs-parent difference is attributable to D1 or D2 by construction — nothing else moved.

Honest class statement (INFERENCE, per IDENTIFIABILITY_MAP §7). Neither quantity enters the scoring functional directly — both pass through frozen decision surfaces — so this lane does not claim class-(a) construction. Ship evidence is G-6(iv): the same component class (split-fidelity) carries a prior board read (W54-C, +0.03158). Honest expectations at build time were D1 −0.0005…+0.0015 and D2 −0.001…+0.002 total-score units — proxy-mediated, sign not guaranteed. The March board read for the floor was +0.00336 over the parent; that is class evidence, not a constant projected to April.

6. Selection protocol: seven gates, then certification

The T1 swap was promoted only after passing every gate (builds/w28_scale_bundle_gates/VERDICT.md, gates.py):

- **G1/G2 — gain and deflated significance.** Blinded Oct–Jan lane

evaluation: ensemble ΔS_state = **+0.00594** vs the previous best, bootstrap 95% CI [+0.00562, +0.00626], **z = 35.7**, far above the 2.67σ bar. Both seeds agree (+0.00595 / +0.00592); all four blinded months positive; worst panel×regime cell +0.00118.

- **G3 — cross-month replication.** Same-sign gains on 4/4 blinded months

plus 10/10 panels on the sealed February selection month — five distinct months of evidence, all positive.

- **G4 — no large regression.** Worst panel×regime cell +0.00118; zero

cells ≤ −0.01.

- **K2 — S_FD adjudication.** The T1 swap changes the physics-consistency

term too: ΔS_FD = +0.00061 (lane) / +0.00246 (Feb), both positive — no hidden drag.

- **K4 — conservation guard.** Relative conservation-residual on matched

Feb panel-days: candidate/production ratio **0.621 ≤ 1.25** (clean bound) — the most conservation-consistent field yet measured; the T3 leg's input is not degraded.

- **G5 — bundle ablation (marginal ≥ 0.002).** Marginal of the T1 swap vs

the previous board leader (0.81458): 0.35×0.00594 + (0.15/3)×0.00061 = **+0.00211** > 0.002. Thin on the point estimate; corroborated by the independent Feb scale-minus-previous read (+0.00602 S_state, tight CI).

- **G7 — sealed February selection.** Pre-declared frozen rule

(ΔS_state_Feb > 0, one run, pre-open hashes): production 0.91765 → candidate **0.93851**, Δ = **+0.02085**, 95% CI [+0.01901, +0.02252], all 10 panels positive.

A regime gate (D3 congestion labels) additionally confirmed the blinded eval months were *more* congestion-heavy than the board months — the regime mismatch runs in the conservative direction. Expected bundle effect vs the previous leader: ≈ +0.0021 total-score units (≈ 0.8167 expected public from 0.81458); the board returned **0.81787** for that candidate.

The floor's own selection (FACT — CAMPAIGN RECORD, W58-L1 audit 2026-10-20; FINAL_SELECTION_LEDGER.md addendum 2026-11-05). The floor was not selected on its March delta. It was certified on composition: class split **6 target-identified / 17 seed-invariant / 12 fitted-residue** pairs; **level discipline PASS** — the bundle's only train-pooled level carriers (pooled priors, pooled occ_ff) are gone, replaced by per-split re-estimates with zero pooled fallback; every other emitted level is pooled seed-invariant

structure or priced residue. The D1/D2 change shipped under G-6(iv) (same component class as W54-C's split-fidelity fix, March +0.03158). Final selection (6 November) is by class composition, not March delta, per the standing selection policy — and the floor is the only candidate that survived it: every challenger since certification was killed or eliminated on board evidence (§7).

Epistemic caveat, disclosed (INFERENCE). K3′ (board delta ≥ +0.010 every month; physics delta ≥ −0.020 every month; same-sign on ≥3 months) was adopted 3 November as a uniform rule *after* the b10 results were known. A uniform post-result rule is not a preregistered rule; its rationale is defensible but it does not erase adaptive decision-making. No further gate recalibration is authorised. The b10 bundle that prompted it scored 0.72109 on the board and was killed the same day.

7. Analysis

Why the scale-up works. The T1 task is dominated by the conditional mean: three independent post-hoc correction families (additive recalibration, congestion-transition classification, variance-weighted spatial correction) were all killed at oracle cost during the campaign — the model's residual field is effectively white noise on the current features, so the conditional mean is already optimal. Gains therefore come from the base model itself: more training rows (3.7M with census-matched augmentation), episode/wave features that expose congestion structure, and pooled priors that stabilize the link-level mean. The D1/D2 fixes extend the same logic to serving time: the level statistics a frozen surface consumes should come from the scored split itself, not from a pooled training month — the disease, measured, is that a split's own month differs from the pooled mean by 2.52 km/h (speed) and 156.3 vph (flow) on average at link×slot level.

Honest negative results — the six doors (CONTRADICTED/FAILED, each with a named mechanism; decision_log.md 2026-11-03/04). The campaign ran 253 registered experiments; the final week closed every remaining route to beating the floor, and each closure is preserved with its mechanism:

1. **C15 — disagreement blend.** KILLED: the two W28 seeds are nearly identical, so inverse-variance weights collapse to uniform (blend Δ −3.15e-07). Mechanism: *arm correlation collapse*.

2. **C5 — J1 genuine-closure refit.** ELIMINATED on board evidence: 0.85191 vs floor 0.85330 (Δ −0.00139) despite a Tier-2 ruler reading +0.01387 on the queue leg — the second ruler-vs-board disagreement in two days. The internal ruler, not the board, was wrong.

3. **RS1 — anomaly-prior retrain (8-day).** KILLED at the monkey gate: the identity arm alone lost −0.11266 — the monkey tested fitting on 8 days, never the prior. Mechanism: *population-shrink confound*.

4. **R1–R4 — blind-team repairs.** KILLED, all four, each with a named mechanism: group-mean offsets inject noise (released-cell means do not predict masked-cell error, corr +0.10); T+30 queue removals delete true positives (the "evidence-free" additions are 96% true); at-T reconciliation trusts a *lagging* row (J1's T−5 anchor agrees with derived truth 97.6% vs the rule's 2.4%); variance restoration achieves its target slope and lowers the score (optimal shrinkage); congested-context pulls smear localized queues (ΔS_state −0.303). The blind team's *patterns* were real; every truth-free *correction direction* was wrong. These are five pre-registered, mechanism-tested confirmations that post-hoc avenues are dead.

5. **C4 — detector-reliability weighting.** KILLED at decision: ceiling +0.001–0.002 sits 5–10× below the K3′ +0.010 bar, so even a pass could not promote; its class had just been measured dead five times.

6. **RS1R — full-population prior swap.** KILLED at Stage 2:

ΔS_state = −0.002603, negative in all regimes — retraining with the interpolated anomaly prior *degrades* the surface vs retraining with D1. Mechanism: *prior-swap dead (C2)*, a directional contradiction, not a null result.

What the kill record measures (INFERENCE). The fitted surfaces (W28/J1/W54-C) are strong, and the only lever that has ever beaten them is retraining — Kill #71 ("only retraining works") now has five independent confirmations and is measured, not assessed. The campaign's final honest statement (§17.1, 2026-11-04) is that no credible route to breaking 0.85330 remained in the experimental program, and no credible route to 0.95 was evidenced. We report that plainly rather than manufacturing a seventh door. The floor's value is therefore not that it is the best imaginable submission, but that it is the best *honestly evidenced* one — and that its reproduction is proven (§8).

Practical value. The floor's prediction disagreement vs production was not priced through the official scorer for this paper (the earlier 0.81787 value bridge — 178,045.7 vehicle-hours at stake — belongs to a superseded candidate and is not carried forward). We do not report unmeasured real-world savings.

8. Reproducibility statement

The organizers' rerun must satisfy **prediction-equality**, not file-hash equality, on every path:

1. **Environment.** requirements.txt pins all 8 packages, verified

against the live build environment; a Dockerfile (python:3.12-slim) reproduces it. (Dockerfile test-build requires a docker host; the pins themselves are venv-verified. The Dockerfile is retained as an untested reference, explicitly dropped from the verified path by a recorded decision — DOCKERFILE_STATUS.md.)

2. **Training data.** build_train_w28_scale.py regenerates the 10

training shards from the raw masked parquets; acceptance is 10/10 shard md5s vs shard_md5.txt. The full rehearsal re-ran this build end-to-end (see the runbook log); re-verified 2026-11-06 — with one honest correction: the first 2026-11-06 check read 5/10 because the Oct-12 rehearsal had rebuilt shards p00–p05 into the live dir with different parquet serialization bytes (content identical). The live shards were restored from the verified backup (stagea_backup/, md5s match the manifest) and re-verified **10/10 OK**; the pre-restore bytes are preserved in live_p00_p05_pre_restore_20261106/. Full record: rehearsal_floor_20261106.log §A2.

3. **Models.** The verbatim Kaggle trainers

(kernel_train_speed_kmh.py, kernel_train_flow_vph.py) retrain all four models from the on-Kaggle dataset. Acceptance: identical train_meta (n_train, best_iteration, val_RMSE to all 16 digits) **and bit-identical predictions on a 2000-row probe** (max|Δ| = 0.0) against the shipped .cbm files. The .cbm **file** md5s are explicitly the wrong test: CatBoost regenerates a random 35-byte save-time UUID on every save_model call (all 1500 trees and parameters byte-identical; documented in HARDENING_REPORT.md §3). The award package's acceptance checks use prediction-equality throughout.

4. **Training-procedure fidelity (REPRODUCED, 2026-11-04).** An

independent full-population retrain (RS1R Stage 1, private GPU kernel, 5/5 fits, 52 min) replicated the W28 training procedure on W28's manifest-exact 320-panel-day population: baseline S_state 0.905826 reproduced to d = 9.95e-09, and the identity gate passed at |ΔS_state| = **0.000524** ≤ 0.002 (determinism noise ε = 0.000082). This is the strongest procedure-fidelity evidence in the package: the pipeline reproduces the champion's level from scratch. (The Stage 2 treatment built on this pipeline was killed on its own merits — §7, door 6 — which does not touch the identity result.)

5. **D1/D2 fix intermediates.** The per-split priors (20 npz, md5s in

builds/w57_d1d2_splitfix/REPORT.md and REPRODUCE_FINAL.sh stage [3]) and per-split occ_ff (occ_ff_by_split.parquet, md5 03919c6af026d66f80e2bad33425e071) ship as frozen, md5-gated inputs. The D1

base run reproduces the parent's state rows (speed 3.6e-15, flow 9.1e-13) and the D2 base run reproduces the shipped emitter files with 0 diffs on 174,000 rows — so the fix intermediates are attributable by construction. A D1 spot re-run (D7_I10_W, both splits) in the 2026-11-06 floor rehearsal reproduced the lane's intermediates md5-identically (validation 1f366572c575b65a1328cc6809e69322, private a50ab607bc3a61aa476eb9636087307a).

6. **Inference.** infer_w28.py mode valpriv regenerates all 20

intermediate state files; acceptance is 20/20 files with unmatched=0 per panel. The 2026-10-12 rehearsal reproduced all 20 from raw inputs.

7. **Merge / assembly.** merge_candidate.py + verify_diff.py

reproduce the W28 submission with ids/tasks identical, queue_pred/ path_flow bit-identical, and per-cell fidelity speed ≤ 1e-14 / flow ≤ 1e-12 vs the shipped file (CSV float round-trip noise only). The floor assembly (builds/w57_d1d2_splitfix/assemble.py) re-ran in the 2026-11-06 floor rehearsal and reproduced the shipped floor file md5-identically (e6db2a22a5e676955ab225358c0e378c).

8. **Frozen artifacts.** The T2 legs (queue_006g_gj1_*.csv) rerun

bit-identically from their script; the T4 leg is the W54-C own-counts refit (odme_candidate_owncounts.csv, md5 aa8081eabff6c98ae762c5f337760602, solver qp_solver.py shipped). The 0.81787 candidate's T4 leg (exp001/odme_all.csv, md5 36d6568f17cacea5205e69d42f35d142) is a frozen artifact whose exact inputs proved unrecoverable after three reconstruction attempts — it is superseded with the floor and recorded here only because earlier drill documents cite it.

The award gate (predictions within 1% of the private score) is cleared by roughly ten orders of magnitude on the T1 leg: the rerun is prediction-identical, not merely within tolerance.

9. Limitations

- The final private month is unobserved by definition; our case rests on

five-month consistency plus the floor's class composition, not on any single number.

- The G5 bundle marginal (+0.00211) is a thin pass on the point estimate;

the February corroboration (+0.00602 S_state, tight CI) is the tighter evidence.

- D1/D2 are **proxy-mediated class (a)**: the statistics enter through

frozen decision surfaces, so no class-(a) uplift is claimed and the sign is not guaranteed. The A1 pseudo-target back-test of the exact D1/D2 procedure is not on file, and D1 carries a view tilt (built from masked *view* cells, not unmasked truth frames). These are disclosed transfer risks, not hidden ones.

- The W54-A onset leg does not ride into April: March priced it at

≈0 (inside the luck floor), and the April assembly reverts onset to the 006G construction. The submitted file carries W54-A onset rows verbatim.

- The competition's top public scores (≈ 0.93 at the 3 November snapshot)

are unexplained headroom. An earlier campaign verdict calling them arithmetically near-impossible is downgraded to hypothesis: top teams may have a method we did not find. We deliberately did not chase them, accepting a lower public number in exchange for private-month robustness — and we do not claim the board is wrong.

10. Conclusion

A 0.85330 floor built on a seven-gate selection protocol, a scaled and census-matched training recipe, split-local serving-time statistics, and a fully rehearsed reproduction path — including an independent full-population

retrain that replicates the champion's level to |ΔS| = 0.000524. The campaign's standing rule — never promote what five months cannot agree on — is the reason this submission, and not a higher-variance alternative, is our final selection. Behind it stands an honest record: 253 experiments, six doors closed with named mechanisms, and a kill record that measures why the floor stands.

Artifact index (award package, `~/workspace/tfb/ops/award_pkg/`)

- requirements.txt, Dockerfile, DOCKERFILE_STATUS.md
- PRIORS_STATEMENT.md (pooled base + §6 D1 per-split addendum)
- HARDENING_REPORT.md, COMPLETION_REPORT.md (corrected 2026-11-06)
- REPRO_DRILL_W28_JOINT.md, REHEARSAL_RUNBOOK.md (+ §7 execution logs)
- EXECUTION_REPORT.md, FROZEN_ARTIFACT_DISCLOSURE.md
- REPRODUCE.sh (W28 one-panel), REPRODUCE_FINAL.sh (floor, identity filled)
- rehearse_floor_20261106.sh, rehearsal_floor_20261106.log (floor rehearsal)
- Prior evidence: ops/REPRO_DRILL_W28_SCALE.md, builds/w28_scale_bundle_gates/VERDICT.md, builds/w28_t1_scale/STATUS.json, builds/w57_d1d2_splitfix/REPORT.md, intel/CERTIFIED_FLOOR_57429904.md, lanes/rs1_resurrect/HANDOFF_STAGE1_16_1.md, lanes/rs1_resurrect/HANDOFF_FINAL_16_1.md